Gazebo Quick Start

DR-xcobo 2.0 | v1.0 | ROS 2 Jazzy / Harmonic | English

Environment: Ubuntu 24.04 LTS Desktop with ROS 2 Jazzy Desktop installed. Use Bash. This package connects Gazebo Harmonic, MoveIt 2 and RViz2.

Download the Gazebo simulation ZIP into **~/Downloads/** and keep its filename. Model, meshes and planning configuration are included; the separate course URDF and MoveIt packages are not required.

1. Install dependencies (once)

```
source /opt/ros/jazzy/setup.bash
sudo apt update
sudo apt install unzip build-essential python3-yaml \
  python3-colcon-common-extensions ros-jazzy-ament-cmake \
  ros-jazzy-ament-cmake-test ros-jazzy-xacro \
  ros-jazzy-moveit ros-jazzy-ros-gz \
  ros-jazzy-gz-ros2-control ros-jazzy-ros2-controllers
```

If the first line cannot find the file, install ROS 2 Jazzy Desktop first. ros-jazzy-ros-gz supplies the matching Gazebo libraries. This package does not use Gazebo Classic.

2. Extract & build (once)

```
mkdir -p ~/dr_xcobo_gazebo_ws/src
cd ~/dr_xcobo_gazebo_ws/src
unzip ~/Downloads/dr-xcobo-2.0-ros2-gazebo-v1.0.zip
cd ~/dr_xcobo_gazebo_ws
colcon build --symlink-install \
  --packages-select dr_xcobo_gazebo
```

Keep the inner package name **dr_xcobo_gazebo**. Build in a terminal sourced only with Jazzy. Use colcon build for this package, not catkin_make.

3. Each new terminal: source & launch

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_gazebo_ws/install/setup.bash
ros2 launch dr_xcobo_gazebo gazebo.launch.py
```

Wait for both Gazebo and RViz2 to open. Simulation starts automatically. Do not start roscore or another MoveIt demo alongside this launcher.

This is a baseline simulation with gravity, a ground plane and a fixed base. It does not connect to a physical robot or add camera images or force sensors.

Planning & Simulation

DR-xcobo 2.0 | Gazebo Harmonic / MoveIt 2 | English

4. Plan in RViz2 & execute in Gazebo

1. Check simulation: Gazebo should display the arm and ground. If paused, press Play first; a paused simulation prevents the clock and controllers from advancing normally.

2. Set a goal in RViz2: select Planning Group **arm** and Start State **current**. Use the interactive marker or Joints tab to choose a nearby, collision-free goal above the ground.

3. Click Plan, then Execute: inspect the preview before execution. Watch the six-axis arm move in Gazebo; RViz2 updates from simulated joint feedback. A separate controller holds both fingers at 0.02 m.

4. Stop: press **Ctrl+C** in the launch terminal. Objects added in Gazebo are not automatically added to the MoveIt collision scene; configure them separately. A successful plan is not a hardware safety guarantee.

Daily use & source

Repeat the two source lines in Step 3 for each new terminal, then launch. The first loads ROS and the second loads this workspace; both affect only that terminal. Routine use normally needs no reinstall, extraction or rebuild.

Optional: add both source lines to **~/.bashrc**, but omit ros2 launch. Do not also auto-source Noetic or unrelated overlays. Rebuild after adding files or changing build configuration.

No motion or inactive controllers

Keep the launcher running. In another terminal, check controllers and the simulation clock:

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_gazebo_ws/install/setup.bash
ros2 control list_controllers
ros2 topic echo /clock --once
```

joint_state_broadcaster, **arm_controller** and **finger_hold_controller** should be **active**. If not, inspect the first launch error and confirm Gazebo is not paused.

Headless launch (optional)

```
ros2 launch dr_xcobo_gazebo gazebo.launch.py \
  gui:=false rviz:=false
```

Stop the existing launch first, then source as usual. Check state with ros2 control list_controllers or ros2 topic echo /joint_states --once. The headless command does not send a motion goal automatically.

Model & verification scope

Original meshes, moving-joint structure, position limits and existing inertias are retained. A fixed base, missing base inertia and simulation control settings are added. Velocity, effort and gains are demonstration values, not hardware ratings. See MIGRATION.md.

Files and configuration have been checked statically. This package has not been built or run in a ROS 2 / Gazebo environment.

Course resources | Official installation | Gazebo control plugin

Gazebo Quick Start

DR-xcobo 2.0 | v1.0 | ROS 2 Jazzy / Harmonic | 中文版

环境：Ubuntu 24.04 LTS 桌面版，已安装 ROS 2 Jazzy Desktop；使用 Bash 终端。本包连接 Gazebo Harmonic、MoveIt 2 与 RViz2。

下载 Gazebo 仿真功能包 ZIP，保留文件名并放在 ~/Downloads/。模型、网格与规划配置已包含；无需另装课程中的独立 URDF 或 MoveIt 包。

1. 安装依赖（首次使用）

```
source /opt/ros/jazzy/setup.bash
sudo apt update
sudo apt install unzip build-essential python3-yaml \
  python3-colcon-common-extensions ros-jazzy-ament-cmake \
  ros-jazzy-ament-cmake-test ros-jazzy-xacro \
  ros-jazzy-moveit ros-jazzy-ros-gz \
  ros-jazzy-gz-ros2-control ros-jazzy-ros2-controllers
```

若第一行找不到文件，请先安装 ROS 2 Jazzy Desktop。ros-jazzy-ros-gz 会安装与 Jazzy 配套的 Gazebo 库；此包不用 Gazebo Classic。

2. 解压并构建（首次使用）

```
mkdir -p ~/dr_xcobo_gazebo_ws/src
cd ~/dr_xcobo_gazebo_ws/src
unzip ~/Downloads/dr-xcobo-2.0-ros2-gazebo-v1.0.zip
cd ~/dr_xcobo_gazebo_ws
colcon build --symlink-install \
  --packages-select dr_xcobo_gazebo
```

保留内部包名 dr_xcobo_gazebo。在只加载 Jazzy 的终端构建；此包使用 colcon build，不使用 catkin_make。

3. 每个新终端：加载环境并启动

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_gazebo_ws/install/setup.bash
ros2 launch dr_xcobo_gazebo gazebo.launch.py
```

等待 Gazebo 和 RViz2 两个窗口出现。启动入口自动开始仿真；不需要另开 roscore 或启动其他 MoveIt 演示。

本包提供带重力、地面与固定基座的基础仿真；不连接真实机械臂，也没有添加相机图像或力传感器。

规划与仿真操作

DR-xcobo 2.0 | Gazebo Harmonic / MoveIt 2 | 中文版

4. 用 RViz2 规划，在 Gazebo 中执行

① **确认仿真运行**：在 Gazebo 中应看到机械臂与地面。若已暂停，先点击 Play；暂停时仿真时钟与控制器不会正常推进。

② **在 RViz2 设置目标**：Planning Group 选择 arm，Start State 选择 current。用交互标记或 Joints 页设置一个近距离、地面上方、无碰撞的目标。

③ **点击 Plan，再点击 Execute**：先检查规划预览，成功后执行。在 Gazebo 中观察六轴运动，RViz2 根据仿真关节反馈同步更新。两个夹指由独立控制器保持在 0.02 m。

④ **结束**：在启动终端按 Ctrl+C。Gazebo 内新增的障碍物不会自动同步到 MoveIt 碰撞场景，需另外配置；不要把规划成功当作真机安全保证。

日常启动与 source

每个新终端都执行上一页第 3 步的两行 source，再启动。第一行加载 ROS，第二行加载本工作空间；它们只对当前终端生效。通常无需再次安装、解压或构建。

可选：将两行 source 加入 ~/.bashrc，不要加入 ros2 launch，也不要同时自动加载 Noetic 或多个不相关工作空间。新增文件或修改构建配置后重新 colcon build。

没有运动或控制器未启动

保留启动窗口；另开终端，确认控制器已激活、仿真时钟正在更新：

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_gazebo_ws/install/setup.bash
ros2 control list_controllers
ros2 topic echo /clock --once
```

应看到 joint_state_broadcaster、arm_controller 与 finger_hold_controller 为 active。若失败，检查启动日志中的首个错误，并确认 Gazebo 未暂停。

无窗口启动（可选）

```
ros2 launch dr_xcobo_gazebo gazebo.launch.py \
  gui:=false rviz:=false
```

先结束已运行的启动进程，再按前述方法 source。没有图形窗口时，可用 ros2 control list_controllers 与 ros2 topic echo /joint_states --once 检查状态；上述命令不会自动发送运动目标。

模型与验证范围

沿用原模型的网格、活动关节结构、位置限位与已有惯性；仅补充固定基座、缺失的基座惯性及仿真控制参数。速度、力矩和控制增益是演示设定，不是设备额定参数。详见 MIGRATION.md。

已完成文件与配置静态核对，尚未在 ROS 2 / Gazebo 环境中构建或运行验证。

课程资源 | 官方安装说明 | Gazebo 控制插件